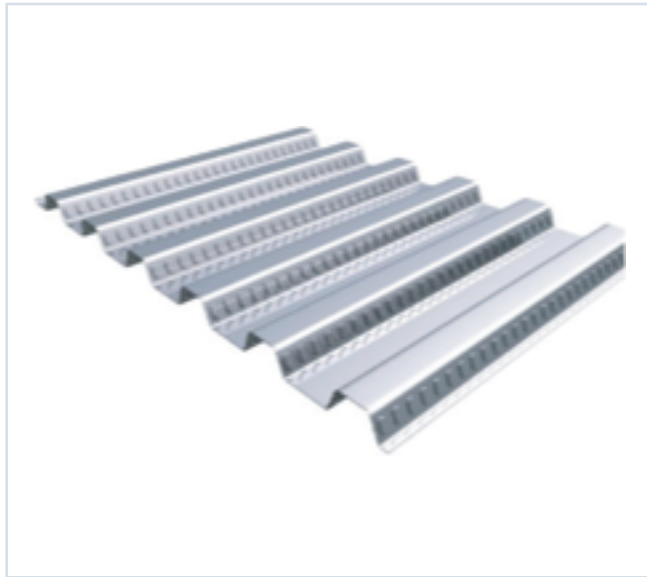




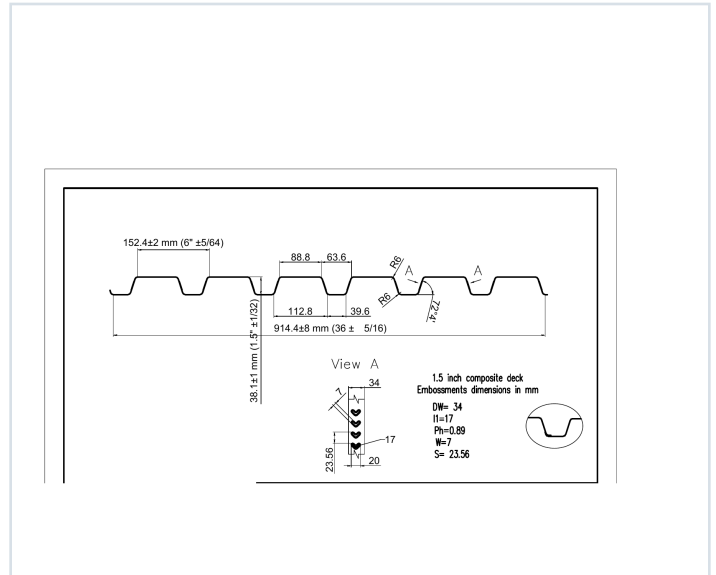
DECKING PRODUCT SUBMITTAL

PRODUCT	GRADE	GAUGE	COATING
1.5" Composite Deck	Grade 50	22 ga	G40

PRODUCT VIEW



PROFILE DRAWING



SECTION PROPERTIES · 22 GAUGE

Thickness (in.)	Weight (psf)	Fy (ksi)	S+ (in³/ft)	S- (in³/ft)	M+ (in-lb/ft)	M- (in-lb/ft)	I+ (in⁴/ft)	I- (in⁴/ft)
0.0295	1.6	50	0.170	0.179	5101	5358	0.144	0.167

SOURCE AND USE

1. Source: Refined Metal Steel Catalog 2025, published pages 45, 46, 47, 48
2. ASD section values above are the exact published row for this grade and gauge.
3. Coating selection does not change the published structural performance values.
4. Verify project-specific loading, fastening, span, concrete, and code requirements with the design professional.

PUBLISHED PERFORMANCE

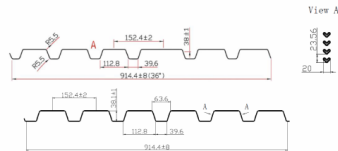
15CD-FY50-22-G40 · 22 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

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STEEL CATALOG PAGE 45

GRADE 50 STEEL



Section Properties

Gage	Design Thickness (inches)	Weight (lb/ft)	F _y (ksi)	F _t (ksi)	S _x (in ³)	S _y (in ³)	ASD (S _x + 147)	ASD (S _y + 147)	I _x (in ⁴)	I _y (in ⁴)
22	0.0295	16	50	0.770	0.222	0.0179	501	558	0.144	0.107
20	0.0368	20	50	0.294	0.330	0.022	607	660	0.182	0.290
18	0.0474	26	50	0.294	0.330	0.022	607	660	0.257	0.290
16	0.0598	33	50	0.294	0.330	0.022	607	660	0.341	0.363

Note: All section properties and ASD/beam strengths are calculated in accordance with ANSI/AISC 360-10, AISC 360-10 and AISC 360-10.

Shear and Web Crippling

Gage	V _u (kips)	Web Crippling (R _u), lbs/ft	Web Crippling (R _u), lbs/ft
		One Flange Loading	Interior Bearing
22	2424	1015	1015
20	3803	1535	1535
18	5052	2077	2077
16	6279	2597	2597

Note: All section properties and ASD/beam strengths are calculated in accordance with ANSI/AISC 360-10, AISC 360-10 and AISC 360-10.

Allowable Uniform Downward Loads, ASD (PSF)

Span	Gage	5'-0"	5'-6"	6'-0"	6'-6"	7'-0"	7'-6"	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"
Single	22	158	12	94	80	69	60	53	47	42	38	34
	20	272	142	100	82	71	62	55	49	44	40	36
	18	235	124	90	75	65	56	49	43	39	35	31
	16	202	108	80	67	58	50	43	38	34	30	27
Double	22	143	107	83	69	59	51	44	39	35	31	27
	20	248	142	100	82	71	62	55	49	44	40	36
	18	215	124	90	75	65	56	49	43	39	35	31
	16	182	108	80	67	58	50	43	38	34	30	27
Triple	22	179	143	107	83	69	59	51	44	39	35	31
	20	292	165	118	96	82	71	62	55	49	44	40
	18	254	139	107	83	69	59	51	44	39	35	31
	16	215	124	90	75	65	56	49	43	39	35	31

Note: For loads that cause L/120 Deflection, multiply by 2.0. For loads that cause L/180 Deflection, multiply by 1.5. For loads that cause L/360 Deflection, multiply by 0.67.

STEEL CATALOG PAGE 46

Span	Gage	5'-0"	5'-6"	6'-0"	6'-6"	7'-0"	7'-6"	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"
Single	22	16	12	94	80	69	60	53	47	42	38	34
	20	16	12	94	80	69	60	53	47	42	38	34
	18	16	12	94	80	69	60	53	47	42	38	34
	16	16	12	94	80	69	60	53	47	42	38	34
Double	22	16	12	94	80	69	60	53	47	42	38	34
	20	230	124	90	75	65	56	49	43	39	35	31
	18	230	124	90	75	65	56	49	43	39	35	31
	16	230	124	90	75	65	56	49	43	39	35	31
Triple	22	16	12	94	80	69	60	53	47	42	38	34
	20	254	139	107	83	69	59	51	44	39	35	31
	18	254	139	107	83	69	59	51	44	39	35	31
	16	254	139	107	83	69	59	51	44	39	35	31

Note: For loads that cause L/120 Deflection, multiply by 2.0. For loads that cause L/180 Deflection, multiply by 1.5. For loads that cause L/360 Deflection, multiply by 0.67.

Construction Span Table - 20 psf Construction Load

Total Slab Depth	Normal Weight Concrete (145 pcf)	Lightweight Concrete (115 pcf)
	Deck Type	Deck Type
	Maximum Unstayed Clear Span	Maximum Unstayed Clear Span
	1 span	1 span
	2 span	2 span
	3 span	3 span
3.0 (8-10) 31 PSF	15x16 ga 8' 0" 7' 0" 7' 0"	15x16 ga 8' 0" 7' 0" 7' 0"
	15x18 ga 9' 0" 8' 0" 8' 0"	15x18 ga 9' 0" 8' 0" 8' 0"
	15x20 ga 10' 0" 9' 0" 9' 0"	15x20 ga 10' 0" 9' 0" 9' 0"
4.00 (12-20) 37 PSF	15x16 ga 8' 0" 7' 0" 7' 0"	15x16 ga 8' 0" 7' 0" 7' 0"
	15x18 ga 9' 0" 8' 0" 8' 0"	15x18 ga 9' 0" 8' 0" 8' 0"
	15x20 ga 10' 0" 9' 0" 9' 0"	15x20 ga 10' 0" 9' 0" 9' 0"
4.50 (14-20) 43 PSF	15x16 ga 8' 0" 7' 0" 7' 0"	15x16 ga 8' 0" 7' 0" 7' 0"
	15x18 ga 9' 0" 8' 0" 8' 0"	15x18 ga 9' 0" 8' 0" 8' 0"
	15x20 ga 10' 0" 9' 0" 9' 0"	15x20 ga 10' 0" 9' 0" 9' 0"
5.00 (16-20) 49 PSF	15x16 ga 8' 0" 7' 0" 7' 0"	15x16 ga 8' 0" 7' 0" 7' 0"
	15x18 ga 9' 0" 8' 0" 8' 0"	15x18 ga 9' 0" 8' 0" 8' 0"
	15x20 ga 10' 0" 9' 0" 9' 0"	15x20 ga 10' 0" 9' 0" 9' 0"
5.50 (18-20) 55 PSF	15x16 ga 8' 0" 7' 0" 7' 0"	15x16 ga 8' 0" 7' 0" 7' 0"
	15x18 ga 9' 0" 8' 0" 8' 0"	15x18 ga 9' 0" 8' 0" 8' 0"
	15x20 ga 10' 0" 9' 0" 9' 0"	15x20 ga 10' 0" 9' 0" 9' 0"
6.00 (19-20) 61 PSF	15x16 ga 8' 0" 7' 0" 7' 0"	15x16 ga 8' 0" 7' 0" 7' 0"
	15x18 ga 9' 0" 8' 0" 8' 0"	15x18 ga 9' 0" 8' 0" 8' 0"
	15x20 ga 10' 0" 9' 0" 9' 0"	15x20 ga 10' 0" 9' 0" 9' 0"

Note: Web crippling and shear have not been accounted for in these tables. Required bearing should be determined based on specific span conditions.

PUBLISHED PERFORMANCE DATA

15CD-FY50-22-G40 · 22 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

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STEEL CATALOG PAGE 47

LSC Deck - Wide Flange Up				ISNV Deck - Wide Flange Down			
50 ksi				50 ksi			
Live Load	20	psf	with concrete	Live Load	20	psf	with concrete
DL	160	psf		DL	160	psf	
Deck Weight	2	psf		Deck Weight	2	psf	
Live Load	50	psf	without concrete	Live Load	50	psf	without concrete
Total Slab Thickness	Deck Gauge	Maximum Cantilever (inches) NWC	Maximum Cantilever (inches) LWC	Total Slab Thickness	Deck Gauge	Maximum Cantilever (inches) NWC	Maximum Cantilever (inches) LWC
3.5	22	36	36	3.5	22	25	25
3.5	20	31	31	3.5	20	30	30
3.5	18	40	40	3.5	18	38	38
3.5	16	47	47	3.5	16	46	46
4	22	36	36	4	22	25	25
4	20	31	31	4	20	30	30
4	18	40	40	4	18	38	38
4	16	47	47	4	16	46	46
4.5	22	36	36	4.5	22	25	25
4.5	20	31	31	4.5	20	30	30
4.5	18	40	40	4.5	18	38	38
4.5	16	47	47	4.5	16	46	46
5	22	36	36	5	22	25	25
5	20	31	31	5	20	30	30
5	18	40	40	5	18	38	38
5	16	47	47	5	16	46	46
5.5	22	36	36	5.5	22	25	25
5.5	20	31	31	5.5	20	30	30
5.5	18	40	40	5.5	18	38	38
5.5	16	47	47	5.5	16	46	46
6	22	36	36	6	22	25	25
6	20	31	31	6	20	30	30
6	18	40	40	6	18	38	38
6	16	47	47	6	16	46	46

Assumes that backspan is greater than the cantilever spans.
Applicable to single, double, triple spans or greater.

Controlled by concentrated load = 30 psf

Controlled by concrete weight = 30 psf

Controlled by Deflection

STEEL CATALOG PAGE 48

22 ga Normalweight Concrete (145 pcf, F'c = 3,000 psi)									
Slab Thickness (inches)	Weight (psf)	8'-0"	9'-0"	10'-0"	11'-0"	12'-0"	13'-0"	14'-0"	15'-0"
3.5	31	400	400	372	334	289	252	210	170
4	37	400	400	400	397	340	298	255	215
4.5	43	400	400	400	400	400	397	331	281
5	49	400	400	400	400	400	400	369	319
5.5	55	400	400	400	400	400	400	400	400
6	61	400	400	400	400	400	400	400	400

Slab Thickness (inches)	8'-0"	9'-0"	10'-0"	11'-0"	12'-0"	13'-0"	14'-0"	15'-0"
3.5	277	156	135	121	110	96	85	80
4	224	197	175	156	139	125	113	102
4.5	273	241	213	190	170	153	138	126
5	323	286	253	226	200	182	164	149
5.5	375	331	294	262	235	211	191	173
6	400	378	336	300	269	242	218	197

20/18/16 ga Normalweight Concrete (145 pcf, F'c = 3,000 psi)									
Slab Thickness (inches)	Weight (psf)	8'-0"	9'-0"	10'-0"	11'-0"	12'-0"	13'-0"	14'-0"	15'-0"
3.5	31	400	400	400	390	337	283	249	214
4	37	400	400	400	400	400	398	332	282
4.5	43	400	400	400	400	400	400	381	331
5	49	400	400	400	400	400	400	400	400
5.5	55	400	400	400	400	400	400	400	400
6	61	400	400	400	400	400	400	400	400

Slab Thickness (inches)	8'-0"	9'-0"	10'-0"	11'-0"	12'-0"	13'-0"	14'-0"	15'-0"
3.5	236	181	170	151	136	122	110	100
4	274	242	215	192	172	155	140	127
4.5	335	296	263	235	215	190	172	156
5	398	362	315	280	251	226	203	186
5.5	400	400	364	325	292	263	238	216
6	400	400	400	372	334	301	272	247

Note
Due to the profile of the embossments, there is no gain in strength for the composite deck-slab when the deck gets thicker than 20 ga. However, the construction square do get longer for 18 and 16 ga deck.